MCQs for unit -2

1. What are the different types of Carry propagator adders?
    a. **Ripple carry adders**
    b. **Carry lookahead adders**
    c. **Prefix adders**
    d. Skip ahead adders
2. What is the delay in the ripple carry adder, given t_{FA} is the delay of 1 full adder?
    a. **N x t_{FA}**
    b. t_{FA}
    c. (N-1) x t_{FA}
    d. None of the above
3. What is the N-bit delay of a prefix adder?
    a. **$t_{PA} = t_{pg} + \log_2 N(t_{pg\text{-}prefix}) + t_{XOR}$**
    b. t_{PA} = N X t_{pg}
    c. $t_{PA} = t_{pg} + \log_2 N$
    d. $t_{PA} = t_{pg} + t_{XOR}$
4. What is the Sum output of a Half Adder?
    a. A + B
    b. A ⊕ B
    c. AB
    d. $\bar{A}B + A\bar{B}$
5. What is the Carry output of a Half Adder?
    a. A + B
    b. AB
    c. A ⊕ B
    d. $A\bar{B}$
6. What is the Sum output of a Full Adder?
    a. A + B + Cin
    b. A ⊕ B ⊕ Cin
    c. AB + Cin
    d. A ⊕ B

7. What is the Carry output of a Full Adder?
    a. A ⊕ B ⊕ Cin
    b. AB + ACin + BCin
    c. AB + Cin
    d. A + B + Cin
8. A Carry-Lookahead Adder improves speed by:
    a. Reducing the number of bits
    b. Using fewer gates
    c. Computing carries in parallel
    d. Removing carries

9. The generate signal for bit i is:
   a. Pi = Ai ⊕ Bi
   b. Gi = AiBi
   c. Gi = Ai + Bi
   d. Gi = Ai − Bi
10. The propagate signal for bit i in the Harris text is:
   a. Pi = AiBi
   b. Pi = Ai + Bi
   c. Pi = Ai ⊕ Bi
   d. Pi = Ai − Bi
11. The carry equation for a Carry-Lookahead Adder is:
   a. Ci = GiPi
   b. Ci = Gi + PiCi-1
   c. Ci = Ai + Bi
   d. Ci = Pi + Ci-1
12. Which adder generally provides the fastest performance for large word sizes?
   a. Ripple Carry Adder
   b. Carry Skip Adder
   c. Prefix Adder
   d. Serial Adder

13. A Carry-Lookahead Adder improves speed by:

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14. **Which circuit performs arithmetic and logical operations?**

   a. Decoder
   b. Multiplexer
   c. ALU
   d. Counter

15. **Logical left shift by one bit is equivalent to:**

   a. Division by 2
   b. Multiplication by 2
   c. Multiplication by 4
   d. No change

16. **A comparator is used to:**

   a. Add two numbers
   b. Multiply two numbers
   c. Determine relative magnitude of two numbers
   d. Shift a number

17. In binary multiplication, each partial product is generated using:

a. XOR gates only
b. AND operations between multiplicand and multiplier bits
c. OR gates only
d. NOT gates only

18. Which adder has delay proportional to $\log_2 N$?

a. Ripple Carry Adder
b. Serial Adder
c. Prefix Adder
d. Half Adder

19. You are designing a 32-bit processor for a low-cost embedded device. The propagation delay of a full adder is 300 ps, and each two-input gate delay is 100 ps. Which adder implementation would you choose if minimizing delay is the primary goal, and what would be the approximate delay?
A. Ripple-Carry Adder, 9.6 ns
B. Carry-Lookahead Adder (4-bit blocks), 3.3 ns
C. Prefix Adder, 1.2 ns
D. Ripple-Carry Adder, 3.3 ns
Answer: C. Prefix Adder, 1.2 ns
Reasoning: Ripple-Carry delay = $32 \times 300 = 9.6$ns. CLA delay = 3.3 ns. Prefix Adder delay = $100 + \log_2(32) \times 200 + 100 = 1.2$ns. Fastest option is Prefix Adder.

20. You are implementing subtraction in an ALU. To compute $Y = A - B$, which hardware configuration is correct?
A. Invert bits of B, add 1, then add to A with Cin = 1
B. Directly subtract B from A using a comparator
C. Use a ripple-carry adder with Cin = 0
D. Use XOR gates to flip A and add B
Answer: A. Invert bits of B, add 1, then add to A with Cin = 1
Reasoning: Two's complement subtraction is performed by inverting B, adding 1, and then adding to A.

21. You need to multiply a binary number 000011_2 by 16 using minimal hardware. Which operation achieves this?
A. Arithmetic Right Shift by 4
B. Logical Left Shift by 4
C. Rotate Left by 4
D. Add 16 repeatedly
Answer: B. Logical Left Shift by 4
Reasoning: Left shift by N bits multiplies the number by 2^N. Here, $000011_2 << 4 = 110000_2 = 48_{10}$.

22. A designer computes A – B and checks sign bit. **Q:** If sign bit = 1, what does it mean?
A) A > B
B) A < B
C) A = B
D) Overflow

Ans: B
Explanation: Negative result → A < B.

23. ALU configured with F2 = 1, F1:0 = 10. **Q:** What operation occurs?
A) A + B
B) A − B
C) A OR B
D) SLT
Ans: B
Explanation: F2 inverts B and adds 1 → subtraction.

24. 11001 LSR 2. Q: Result?
A) 00110
B) 11110
C) 01110
D) 00100
Ans: A
Explanation: Logical shift right fills with 0s → 00110.

25. In two's complement addition, A = 0111 (7), B = 0100 (4). **Q:** What is the overflow flag?
A) 0
B) 1
C) Depends on carry-in
D) Undefined
Ans: B
Explanation: 7 + 4 = 11 → binary 1011 (negative in 4-bit signed). Overflow occurs when adding two positives yields a negative.

26. A 32 × 32 multiplier is implemented. **Q:** What is product size?
A) 32 bits
B) 64 bits
C) 128 bits
D) 16 bits
Ans: B
Explanation: N × N multiplier → 2N-bit result → 64 bits.

27. A 16-bit logical shifter is built. **Q:** How many multiplexers are required?
A) 16 × 16
B) 16 × 4
C) 16 × $\log_2 16$
D) 16 × 1
Ans: A
Explanation: Each bit requires a 16:1 multiplexer → 16 × 16.

28. A CPU designer chooses between CLA and Prefix adder for a 128-bit ALU. **Q:** Why might prefix adder be chosen despite higher hardware cost?
A) Lower power consumption
B) Logarithmic delay scaling
C) Smaller area footprint
D) Easier HDL coding
Ans: B
Explanation: Prefix adder delay ∝ $\log_2 N$, critical for wide adders in CPUs.

29. A 64-bit ripple-carry adder has tFA = 250 ps. A 64-bit CLA with 8-bit blocks has tpg = 100 ps, tpg_block = 800 ps, tAND_OR = 200 ps, and tFA = 250 ps. **Q:** Which design is faster and why?
    A) Ripple-carry, because fewer gates
    B) CLA, because carry computed in parallel
    C) Ripple-carry, because linear delay is smaller
    D) Both equal
    Ans: B
    Explanation: Ripple delay = 64 × 250 ps = 16 ns. CLA delay ≈ 100 + 800 + (64/8 − 1) × 200 + (8 × 250) = 3.9 ns. CLA is much faster.

30. A = 25, B = 32, F2:0 = 111. **Q:** What is Y?
    A) 0
    B) 1
    C) 25
    D) 32
    Ans: B
    Explanation: SLT → Y = 1 if A < B.

31. A 4-bit comparator checks if A = B. **Q:** Which gate is used per bit?
    A) XOR
    B) XNOR
    C) AND
    D) OR
    Ans: B
    Explanation: XNOR checks equality per bit.

32. To compute A − B, the designer inverts B and adds 1. **Q:** Which input must be set in CPA?
    A) Cin = 0
    B) Cin = 1
    C) Cout = 1
    D) None
    Ans: B
    Explanation: Subtraction = A + (¬B + 1) → Cin = 1.

33. A 32-bit CLA with 4-bit blocks is designed. **Q:** Why is it faster than ripple-carry?
    A) Uses fewer gates
    B) Computes carry in parallel using generate/propagate
    C) Eliminates carry completely
    D) Uses XOR only
    Ans: B
    Explanation: CLA precomputes carry using G and P signals.

34. A circuit designer wants to add two 1-bit numbers but doesn't care about carry-in. **Q:** Which outputs does the half adder produce?
    A) Sum only
    B) Carry only
    C) Sum and Carry-out
    D) Sum, Carry-in, Carry-out
    Ans: C
    Explanation: Half adder → S = A ⊕ B, Cout = A·B. No Cin input.

35. In two’s complement addition, A = 0111 (7), B = 0100 (4). **Q:** What is the overflow flag?

A) 0

B) 1

C) Depends on carry-in

D) Undefined

Ans: B

Explanation: 7 + 4 = 11 → binary 1011 (negative in 4-bit signed). Overflow occurs when adding two positives yields a negative.